Formal Proposal Form

Student Name	Syed Arzanish Shumak Pirzada
Student ID Number	M01039341
Study Status	Full time student
Campus	Dubai Campus
Date of Submission	06/02/2026

Type of Research Proposal	This research proposal is a **design-based and experimental prototyping proposal**. It focuses on the development, fabrication, and control of a low-cost, electromechanical knee exoskeleton designed to assist human mobility. The project explores the feasibility of replacing expensive pneumatic actuators with accessible DC motor technology while maintaining effective gait assistance through sensor fusion algorithms.
Name of Supervisor	Dr Judhi Prasetyo
Title	Mynee – A Cost-Effective Electromechanical Knee Exoskeleton for Mobility Assistance

Key Words	Wearable robotics, exoskeleton, gait analysis, assistive technology, sensor fusion, PID control, electromechanical actuation.
Problem Definition	**1.** Mobility impairments affecting the knee joint (e.g., osteoarthritis, quadriceps weakness) significantly impact quality of life. Current commercial solutions, such as the Roam Robotics Ascend, utilize advanced pneumatic systems that are often prohibitively expensive and complex to maintain for general use. **2.** While high-end exoskeletons exist, there is a significant gap in "accessible robotics"—devices that can be manufactured and maintained using standard electronic components. The problem addressed in this project is the lack of a low-cost, modular knee exoskeleton that provides active torque assistance without the need for heavy air compressors or industrial-grade servo motors.
Research Question(s)	**1.** How can a cost-effective electromechanical system (using DC gear motors) replicate the compliance and assistance of a pneumatic knee exoskeleton? **2.** How can low-cost sensors (IMU and FSR) be utilized to accurately detect user intent and gait phase (e.g., Sit-to-Stand, Walking) in real-time?

Study Area Review	This research is situated within **Biomedical Engineering and Robotics**, specifically focusing on Human-Machine Interaction (HMI). Previous studies have largely focused on rigid, high-torque industrial exoskeletons or soft pneumatic suits. This project builds upon the "Quasi-Direct Drive" and "Assist-As-Needed" control strategies found in recent literature, adapting them for a budget-constrained environment using readily available microcontroller platforms (Arduino/ESP32) and standard planetary gear motors.
Aims	The aim of this project is to design, prototype, and evaluate a budget-friendly knee exoskeleton that provides active torque assistance to the user during walking and standing, demonstrating the viability of low-cost components in medical robotics.
Objectives	• **Mechanical Design & Fabrication:** To design and 3D print (using PETG/PLA+) a lightweight, semi-rigid knee exoskeleton frame that aligns with the user's biological knee joint, incorporating adjustable mounting points for the DC planetary gear motor. • **Actuation System Development:** To implement a cost-effective actuation system using a **PG36-555 Planetary Gear Motor** and an **IBT-2 (BTS7960) High-Current H-Bridge driver**, capable of delivering sufficient torque (approx. 5-10 Nm) to assist in the sit-to-stand motion. • **Sensor Fusion Implementation:** To develop a real-time intent recognition system using a **Kalman Filter** algorithm on an Arduino/ESP32 microcontroller. This will fuse data from **MPU6050 accelerometers/gyroscopes** (angle detection) and **FSRs** (Force Sensitive Resistors) to accurately identify gait phases (Stance vs. Swing). • **Control Algorithm Design:** To program a **PID (Proportional-Integral-Derivative) controller** that regulates the motor's speed and torque output, ensuring smooth "Assist-As-Needed" interaction without sudden jerks or resistance during the swing phase. • **Safety & Validation:** To integrate software hard-stops and physical limit switches to prevent hyperextension (>180 degrees) and to experimentally validate the system's response time (target <100ms latency) and power consumption under load.
Evidence of Requirements	**Availability of Components:** The core actuator (PG36-555) is a standard industrial component available in local UAE robotics markets (e.g., Naif Electronics Market, Besomi Electronics) and online. The microcontroller (Arduino Mega/ESP32) and sensors (MPU6050) are widely supported open-source hardware. **Manufacturing Capabilities:** The university's engineering labs provide access to FDM 3D printers for custom frame fabrication and soldering stations for PCB integration. **Rising Osteoarthritis Rates:** According to the World Health Organization (2024), osteoarthritis is one of the leading causes of disability globally, with knee osteoarthritis affecting millions. Existing commercial solutions like the *Roam Robotics Ascend* cost thousands of dollars, creating a clear "requirement" for low-cost alternatives. **Gap in "Accessible" Robotics:** There is a lack of open-source, affordable exoskeleton designs that can be replicated by researchers in developing regions. This project fulfills the requirement for "frugal innovation" in biomedical engineering. The project requires strict safety protocols. The design incorporates **mechanical hard-stops** (to physically prevent the knee from bending backward) and an **Emergency Kill Switch** accessible to the user, satisfying the ethical requirements for testing wearable devices.

Context Description	This project is conducted within the context of **assistive prototyping**. The system is intended as a "Proof of Concept" (PoC) for educational and research purposes, not as a certified medical device. Constraints include a limited budget (approx. AED 1,000–1,500), reliance on 3D printed structural parts, and safety limits imposed by the university environment (e.g., software and mechanical hard-stops).
Research Methodology	1. **Mechanical Design:** 3D modeling of the frame and motor mounts using CAD software 2. **Electronics Integration:** Interfacing the PG36 motors, IBT-2 drivers, and MPU6050 sensors with the microcontroller. 3. **Algorithm Development:** Coding the PID control loop and Gait State Machine in C++ (Arduino IDE). 4. **Testing & Tuning:** Bench testing the motor response, followed by wearable testing to tune the assistance timing. 5. **Evaluation:** Measuring the system's ability to track the user's movement and provide torque without lag.
Brief Product Description	**Mynee** is a cost-effective, electromechanical knee exoskeleton designed to assist individuals with reduced muscle strength and mobility impairments. Unlike commercial devices that rely on complex and expensive pneumatic systems, **Mynee** utilizes high-torque DC planetary gear motors and an intelligent "Assist-as-Needed" control algorithm. By continuously monitoring the user's limb position via a sensor fusion network (IMUs and FSRs), the system provides synchronized torque support during critical movement phases—specifically "Sit-to-Stand" and walking—thereby reducing the physical effort required by the user while remaining affordable and accessible for general use.
Deliverables	• Approved Research Proposal • System Architecture & Wiring Diagrams • 3D Printed Mechanical Prototype (Wearable) • Control Software (C++ Code) • Final Technical Report & Demonstration Video

Milestones	**1. Proposal & Scope:** Finalizing component list and budget approval. **2. Design Phase:** Completion of CAD models for the leg brace and motor mounts. **3. Subsystem Testing:** successfully reading accurate angles from IMUs and spinning motors via code. **4. Integration:** Assembling the electronics onto the mechanical frame. **5. Control Tuning:** Calibrating the PID loop so the robot moves smoothly with the user. **6. Final Review:** Demonstration of the working prototype and submission of the report.
Resources	• **CAD:** Autodesk Fusion 360 or SolidWorks • **Coding:** Arduino IDE, VS Code (PlatformIO) • **Simulation:** Tinkercad / Proteus (for circuit logic) • **Data Analysis:** Python (Matplotlib) for visualizing sensor logs • **Microcontroller:** Arduino Mega or ESP32 • **Actuators:** 2x PG36-555 Planetary Gear Motors (12V) • **Drivers:** 2x IBT-2 High Current H-Bridges • **Sensors:** 2x MPU6050 IMUs, 2x FSRs • **Power:** 3S LiPo Battery (11.1V) • **Materials:** PETG Filament, Velcro Straps, Aluminum flat bars

Bibliography	1. **Dollar, A. M., & Herr, H.** (2008). Lower Extremity Exoskeletons and Active Orthoses: Challenges and State-of-the-Art. *IEEE Transactions on Robotics*, 24(1), 144–158. (The "Bible" of exoskeleton research). 2. **Lee, H., Kim, W., Han, J., & Han, C.** (2017). The Design of a Quasi-Passive Knee Exoskeleton to Assist During Standing Up. *International Journal of Precision Engineering and Manufacturing*, 18(11), 1595–1602. 3. **Hussain, S., Xie, S. Q., & Jamwal, P. K.** (2013). Control of a Robotic Orthosis for Gait Rehabilitation. *Robotics and Autonomous Systems*, 61(9), 911–919. (Good for explaining your PID control). 4. **World Health Organization.** (2024). *Global Burden of Disease Study: Osteoarthritis*. Retrieved from who.int. **Technical Datasheets (Crucial for Engineering Projects):** 5. **InvenSense Inc.** (2013). *MPU-6000 and MPU-6050 Product Specification Revision 3.4*. Sunnyvale, CA: InvenSense. 6. **Infineon Technologies.** (2010). *BTS7960 High Current PN Half Bridge Datasheet*. Munich, Germany. (This is the datasheet for your IBT-2 Motor Driver). 7. **Dongguan Silence Motor Co.** (2022). *PG36-555 Planetary Gear Motor Technical Specifications*. (Standard reference for your specific motor).
To be completed by the supervisor: Is the Proposal Acceptable?	
If no, please give feedback:	